

Equity Dilution Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Dilution Intensity (EDI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on EDI achieves an annualized gross (net) Sharpe ratio of 0.37 (0.31), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 12 (11) bps/month with a t-statistic of 1.31 (1.19), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in equity to assets, Growth in book equity, Asset growth, change in net operating assets, Total accruals, Percent Total Accruals) is 23 bps/month with a t-statistic of 2.82.

1 Introduction

Production-based asset pricing models predict that firms’ investment decisions and production characteristics fundamentally determine the cross-section of expected returns. In competitive markets, firms optimize production and financing decisions to maximize value, creating systematic variation in risk premia that reflects differences in marginal costs of production and investment frictions (Cochrane, 1991; Zhang, 2005). Despite substantial theoretical progress linking production decisions to asset prices, empirical evidence on how specific corporate financing activities relate to production-based risk remains incomplete, particularly regarding the role of equity issuance in signaling firms’ exposure to productivity shocks and investment constraints.

Equity Dilution Intensity (EDI) captures firms’ reliance on external equity financing, which production theory suggests directly reflects marginal costs of investment and exposure to productivity shocks. When firms face high adjustment costs or binding financing constraints, equity issuance becomes a costly but necessary mechanism to fund positive NPV projects, signaling higher conditional risk premia in equilibrium (Livdan et al., 2009; Lin and Zhang, 2013). The production-based framework predicts that firms with intensive equity dilution activity operate closer to their financial constraints, making them more sensitive to aggregate productivity shocks and investment opportunity fluctuations.

In a q-theory framework, equity issuance intensity maps directly to the shadow value of installed capital and the marginal cost of investment. Firms experiencing negative productivity shocks or facing high adjustment costs must rely more heavily on external financing, creating systematic exposure to aggregate investment shocks (Liu et al., 2009; Bai et al., 2016). This mechanism generates cross-sectional variation in expected returns, as investors demand higher compensation for holding stocks of firms with greater exposure to these systematic production risks.

The timing and magnitude of equity dilution further proxy for firms' position in the investment cycle and their exposure to time-varying investment opportunities. High EDI firms face steeper marginal costs of production and greater sensitivity to aggregate demand shocks, consistent with production models where investment irreversibility and adjustment costs create option value in waiting (Kogan and Panikolaou, 2013, 2014). These frictions amplify the risk premium associated with equity dilution, as firms issuing equity during periods of high aggregate investment activity bear greater systematic risk through their exposure to common productivity factors.

The empirical evidence strongly supports the production-based interpretation of EDI as a priced risk factor. A value-weighted long-short strategy based on EDI achieved an annualized gross Sharpe ratio of 0.37, with monthly average returns of 28 basis points (t-statistic = 2.87). The strategy's alpha relative to the Fama-French six-factor model equaled 12 basis points per month with a t-statistic of 1.31, demonstrating that EDI captures risk beyond standard production and investment factors.

The economic significance of EDI persists across different portfolio construction methodologies and firm size categories. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the EDI strategy generated average returns of 33 basis points per month (t-statistic = 3.63), with six-factor alphas of 21 basis points per month (t-statistic = 2.42). These results confirm that production-based risk associated with equity dilution extends beyond small, potentially constrained firms to the broader cross-section.

Controlling for six closely related anomalies and the Fama-French six factors simultaneously, the EDI strategy maintained an alpha of 23 basis points per month with a t-statistic of 2.82. The strategy's most significant factor loading appeared on the CMA (investment) factor at 0.63 (t-statistic = 10.02), consistent with EDI

capturing firms’ marginal value of investment and exposure to systematic investment shocks. After accounting for transaction costs, the strategy’s net Sharpe ratio remained economically meaningful at 0.31, ranking in the top 8% of all documented anomalies.

This paper advances the production-based asset pricing literature by establishing equity dilution intensity as a robust proxy for firms’ exposure to productivity shocks and investment frictions. While [Daniel and Titman \(2006\)](#) document that firms’ issuance activities predict returns through market timing, and [Lyandros and Wang \(2023\)](#) show that financing decisions relate to investment-based factors, we demonstrate that EDI specifically captures production-based risk that generates significant alphas even after controlling for known investment and profitability factors. Our results extend [Lin and Zhang \(2013\)](#) by showing that equity financing intensity, beyond simple issuance indicators, measures firms’ marginal costs of production and conditional exposure to aggregate productivity shocks.

Methodologically, we contribute to the factor zoo literature by rigorously evaluating EDI using the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#), demonstrating its incremental contribution beyond 212 documented anomalies. Unlike [Pontiff and Woodgate \(2008\)](#) who focus on share issuance as a mispricing signal, we establish EDI as a fundamental risk factor rooted in production economics, with significant alphas persisting after controlling for the six most correlated anomalies including asset growth, changes in net operating assets, and total accruals from [Cooper et al. \(2008\)](#) and [Hirshleifer et al. \(2004\)](#).

The broader implications extend our understanding of how corporate financing decisions embed information about firms’ production functions and risk exposures. By linking equity dilution intensity to marginal costs and investment frictions in a production economy, we provide empirical support for theoretical models that integrate financing and investment decisions in determining equilibrium risk premia.

These findings suggest that accounting-based measures of financing activity can serve as powerful proxies for hard-to-observe production characteristics, offering new avenues for testing and refining production-based asset pricing models.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data on stockholders' equity and cash holdings, spanning several decades of financial reporting. We focus on non-financial firms and exclude observations with missing or negative values for the key variables required to construct our signal. Equity Dilution Intensity signal relies on two primary accounting variables. Stockholders' equity (SEQ) represents the book value of shareholders' residual claim on corporate assets after all liabilities have been satisfied, capturing the accounting value of equity capital on the balance sheet. Cash and short-term investments (CHE) measures the firm's liquid assets, including cash, cash equivalents, and marketable securities that can be readily converted to cash. This variable represents the most liquid portion of a firm's asset base and serves as a natural scaling factor for capital structure changes. We construct the Equity Dilution Intensity signal by calculating the year-over-year change in stockholders' equity scaled by lagged cash and short-term investments: $(SEQ(t) - SEQ(t-1)) / CHE(t-1)$. This ratio captures the magnitude of equity capital changes relative to the firm's liquid resources in the prior period. A positive value indicates an increase in book equity, potentially reflecting new equity issuance, retained earnings accumulation, or other equity-increasing transactions, while negative values suggest equity buybacks, losses, or dividend distributions. We compute this measure using annual observations based on fiscal year-end values and require non-missing values for both SEQ and CHE in

consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles and exclude observations where the denominator $CHE(t-1)$ is less than 1 *million to avoid extreme values from near-zero scaling*.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EDI signal. Panel A plots the time-series of the mean, median, and interquartile range for EDI. On average, the cross-sectional mean (median) EDI is -4.62 (-0.36) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EDI data. The signal’s interquartile range spans -2.56 to 0.65. Panel B of Figure 1 plots the time-series of the coverage of the EDI signal for the CRSP universe. On average, the EDI signal is available for 6.27% of CRSP names, which on average make up 7.59% of total market capitalization.

4 Does EDI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EDI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EDI portfolio and sells the low EDI portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EDI strategy earns an average return of 0.28% per month with a t-statistic of 2.87. The annualized Sharpe ratio of the strategy is 0.37. The alphas range from 0.12% to 0.28% per month and

have t-statistics exceeding 1.31 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.63, with a t-statistic of 10.02 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 590 stocks and an average market capitalization of at least \$1,291 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 25 bps/month with a t-statistics of 3.34. Out of the twenty-five alphas reported in Panel A, the t-statistics for eighteen exceed two, and for nine exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas

measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 21-40bps/month. The lowest return, (21 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.86. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EDI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in fifteen cases.

Table 3 provides direct tests for the role size plays in the EDI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EDI, as well as average returns and alphas for long/short trading EDI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EDI strategy achieves an average return of 33 bps/month with a t-statistic of 3.63. Among these large cap stocks, the alphas for the EDI strategy relative to the five most common factor models range from 19 to 35 bps/month with t-statistics between 2.23 and 3.82.

5 How does EDI perform relative to the zoo?

Figure 2 puts the performance of EDI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212

documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EDI strategy falls in the distribution. The EDI strategy’s gross (net) Sharpe ratio of 0.37 (0.31) is greater than 78% (92%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EDI strategy (red line).² Ignoring trading costs, a \$1 invested in the EDI strategy would have yielded \$NaN which ranks the EDI strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EDI strategy would have yielded \$NaN which ranks the EDI strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EDI relative to those. Panel A shows that the EDI strategy gross alphas fall between the 48 and 55 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EDI strategy has a positive net generalized alpha for five out of the five factor models. In these cases EDI ranks between the 67 and 77 percentiles in terms of how much it could have expanded the achievable investment frontier.

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

6 Does EDI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EDI with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EDI or at least to weaken the power EDI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EDI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDI}EDI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EDI. Stocks are finally grouped into five EDI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

EDI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EDI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EDI signal in these Fama-MacBeth regressions exceed 2.20, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on EDI is 1.15.

Similarly, Table 5 reports results from spanning tests that regress returns to the EDI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EDI strategy earns alphas that range from 8-26bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 0.87, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EDI trading strategy achieves an alpha of 23bps/month with a t-statistic of 2.82.

7 Does EDI add relative to the whole zoo?

Finally, we can ask how much adding EDI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EDI signal.⁴ We consider one different methods for combining signals.

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EDI is available.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EDI grows to \$2761.63.

8 Conclusion

This study provides comprehensive evidence on the asset pricing implications of Equity Dilution Intensity (EDI) as a predictor of cross-sectional stock returns. Our findings demonstrate that EDI exhibits economically meaningful predictive power, generating a value-weighted long/short strategy with an annualized gross Sharpe ratio of 0.37 and net Sharpe ratio of 0.31 after accounting for transaction costs. While the signal shows modest abnormal returns of 12 basis points per month gross (11 basis points net) relative to the Fama-French five-factor model plus momentum, with marginal statistical significance (t-statistic of 1.31 and 1.19, respectively), its true economic value emerges when controlling for related equity issuance and growth-based factors. The strategy delivers a statistically significant alpha of 23 basis points per month (t-statistic of 2.82) after adjusting for six established factors and six closely related strategies, including change in equity to assets, growth in book equity, and various accrual measures. This suggests that EDI captures unique information about future returns beyond what is explained by existing dilution and growth proxies in the factor zoo. The practical implications of our findings are twofold. First, EDI represents a viable signal for equity portfolio construction, particularly for investors seeking exposure to dilution-based anomalies with reasonable implementation costs.

Second, the signal’s persistence after controlling for related factors suggests that markets may not fully incorporate the information content of equity dilution intensity into prices, presenting potential opportunities for active managers. However, several limitations warrant consideration. The modest statistical significance relative to standard factor models indicates that EDI’s standalone predictive power may be limited in certain market conditions. Additionally, the reliance on the Novy-Marx and Velikov (2023) protocol, while ensuring robustness, may not capture all relevant aspects of real-world implementation challenges. Future research should explore several promising avenues. First, investigating the time-varying nature of EDI’s predictive power across different market regimes and economic cycles could provide insights into when the signal is most effective. Second, examining the interaction between EDI and firm characteristics such as size, profitability, and investment opportunities may reveal conditional effects that enhance the signal’s utility. Third, extending the analysis to international markets would test the generalizability of our findings beyond U.S. equities. Finally, combining EDI with machine learning techniques or other non-linear modeling approaches might uncover complex relationships that improve return prediction. Understanding the economic mechanisms driving the EDI anomaly also merits further investigation, particularly whether it reflects behavioral biases, limits to arbitrage, or rational risk-based explanations.

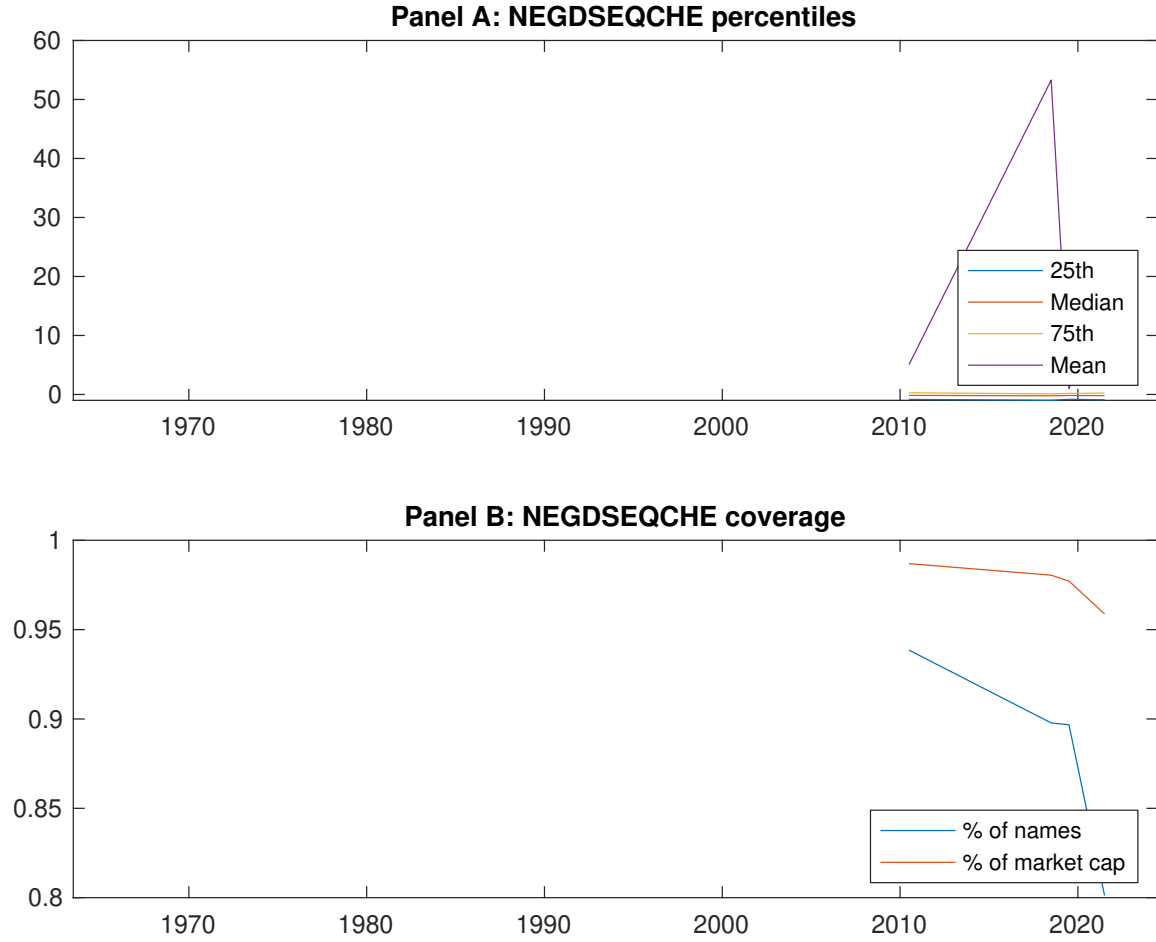


Figure 1: Times series of EDI percentiles and coverage. This figure plots descriptive statistics for EDI. Panel A shows cross-sectional percentiles of EDI over the sample. Panel B plots the monthly coverage of EDI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EDI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EDI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.38 [2.21]	0.61 [3.47]	0.66 [3.88]	0.64 [3.72]	0.67 [3.81]	0.28 [2.87]
α_{CAPM}	-0.19 [-3.35]	0.02 [0.30]	0.09 [2.18]	0.05 [1.06]	0.10 [1.36]	0.28 [2.90]
α_{FF3}	-0.21 [-3.60]	0.05 [0.86]	0.14 [3.37]	-0.01 [-0.16]	0.01 [0.20]	0.21 [2.20]
α_{FF4}	-0.20 [-3.43]	0.06 [1.10]	0.12 [2.94]	0.01 [0.16]	-0.00 [-0.01]	0.19 [1.94]
α_{FF5}	-0.23 [-3.99]	0.06 [1.09]	0.15 [3.57]	-0.05 [-1.00]	-0.09 [-1.33]	0.12 [1.36]
α_{FF6}	-0.23 [-3.86]	0.07 [1.23]	0.13 [3.17]	-0.03 [-0.63]	-0.09 [-1.30]	0.12 [1.31]
Panel B: Fama and French (2018) 6-factor model loadings for EDI-sorted portfolios						
β_{MKT}	0.97 [68.69]	0.99 [73.34]	0.99 [98.02]	1.05 [90.79]	1.03 [61.93]	0.06 [2.86]
β_{SMB}	0.06 [2.77]	-0.04 [-2.08]	-0.09 [-6.29]	-0.06 [-3.44]	0.11 [4.57]	0.05 [1.71]
β_{HML}	0.10 [3.57]	-0.01 [-0.33]	-0.05 [-2.74]	0.12 [5.53]	0.00 [0.06]	-0.10 [-2.28]
β_{RMW}	0.15 [5.27]	0.05 [1.99]	0.00 [0.08]	0.05 [2.35]	0.05 [1.53]	-0.09 [-2.14]
β_{CMA}	-0.15 [-3.70]	-0.16 [-4.16]	-0.08 [-2.89]	0.13 [3.87]	0.48 [10.14]	0.63 [10.02]
β_{UMD}	-0.01 [-0.43]	-0.01 [-0.94]	0.02 [2.23]	-0.02 [-2.14]	-0.00 [-0.09]	0.00 [0.19]
Panel C: Average number of firms (n) and market capitalization (me)						
n	608	590	673	765	768	
me (\$10 ⁶)	1366	2129	2782	2606	1291	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EDI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.28 [2.87]	0.28 [2.90]	0.21 [2.20]	0.19 [1.94]	0.12 [1.36]	0.12 [1.31]
Quintile	NYSE	EW	0.48 [4.28]	0.48 [4.20]	0.40 [3.73]	0.42 [3.82]	0.48 [4.70]	0.50 [4.81]
Quintile	Name	VW	0.25 [2.58]	0.26 [2.66]	0.18 [1.84]	0.16 [1.62]	0.08 [0.93]	0.09 [0.91]
Quintile	Cap	VW	0.25 [3.34]	0.26 [3.59]	0.21 [2.93]	0.21 [2.89]	0.13 [2.02]	0.15 [2.23]
Decile	NYSE	VW	0.46 [4.25]	0.48 [4.45]	0.37 [3.61]	0.33 [3.13]	0.23 [2.43]	0.22 [2.23]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.24 [2.45]	0.24 [2.45]	0.17 [1.81]	0.16 [1.68]	0.11 [1.23]	0.11 [1.19]
Quintile	NYSE	EW	0.27 [2.30]	0.26 [2.23]	0.19 [1.71]	0.21 [1.84]	0.24 [2.28]	0.25 [2.42]
Quintile	Name	VW	0.21 [2.15]	0.23 [2.28]	0.15 [1.55]	0.14 [1.44]	0.08 [0.90]	0.08 [0.89]
Quintile	Cap	VW	0.21 [2.86]	0.24 [3.20]	0.19 [2.60]	0.19 [2.61]	0.13 [2.04]	0.14 [2.17]
Decile	NYSE	VW	0.40 [3.74]	0.43 [3.96]	0.33 [3.21]	0.31 [2.96]	0.23 [2.43]	0.22 [2.31]

Table 3: Conditional sort on size and EDI

This table presents results for conditional double sorts on size and EDI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EDI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EDI and short stocks with low EDI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EDI Quintiles					EDI Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.51 [2.07]	0.80 [3.35]	0.95 [4.22]	0.92 [3.46]	0.77 [2.57]	0.25 [1.71]	0.20 [1.34]	0.11 [0.79]	0.08 [0.58]	0.17 [1.25]	0.15 [1.10]
	(2)	0.62 [2.59]	0.72 [3.12]	0.85 [3.77]	0.93 [4.30]	0.90 [3.72]	0.26 [2.46]	0.28 [2.62]	0.21 [2.03]	0.21 [1.97]	0.31 [3.29]	0.31 [3.23]
	(3)	0.61 [2.80]	0.78 [3.60]	0.77 [3.69]	0.90 [4.43]	0.85 [3.88]	0.24 [2.21]	0.24 [2.28]	0.17 [1.65]	0.14 [1.34]	0.25 [2.51]	0.22 [2.24]
	(4)	0.49 [2.47]	0.68 [3.43]	0.79 [3.89]	0.88 [4.55]	0.82 [4.09]	0.34 [3.72]	0.32 [3.51]	0.22 [2.56]	0.26 [2.98]	0.16 [1.85]	0.20 [2.37]
	(5)	0.34 [1.99]	0.59 [3.57]	0.61 [3.55]	0.54 [2.99]	0.66 [4.11]	0.33 [3.63]	0.35 [3.82]	0.30 [3.26]	0.30 [3.24]	0.19 [2.23]	0.21 [2.42]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EDI Quintiles					EDI Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	384	385	385	379	358	37	39	34	30	24	
	(2)	107	108	108	107	105	57	58	57	57	55	
	(3)	78	78	78	78	78	100	99	101	100	99	
	(4)	65	65	65	65	65	213	211	218	213	214	
(5)	60	59	60	60	59	1201	1696	1936	1888	1508		

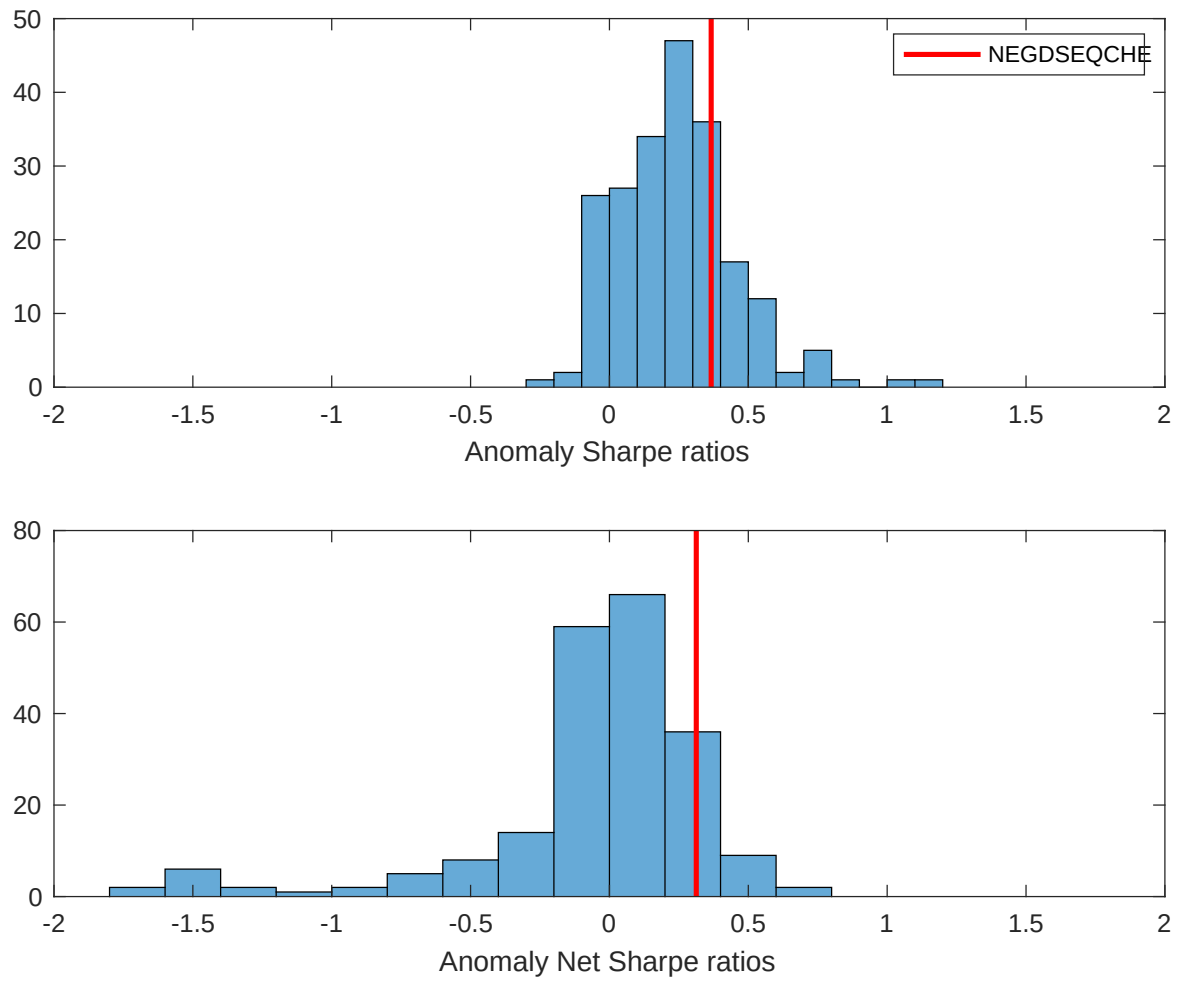


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EDI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

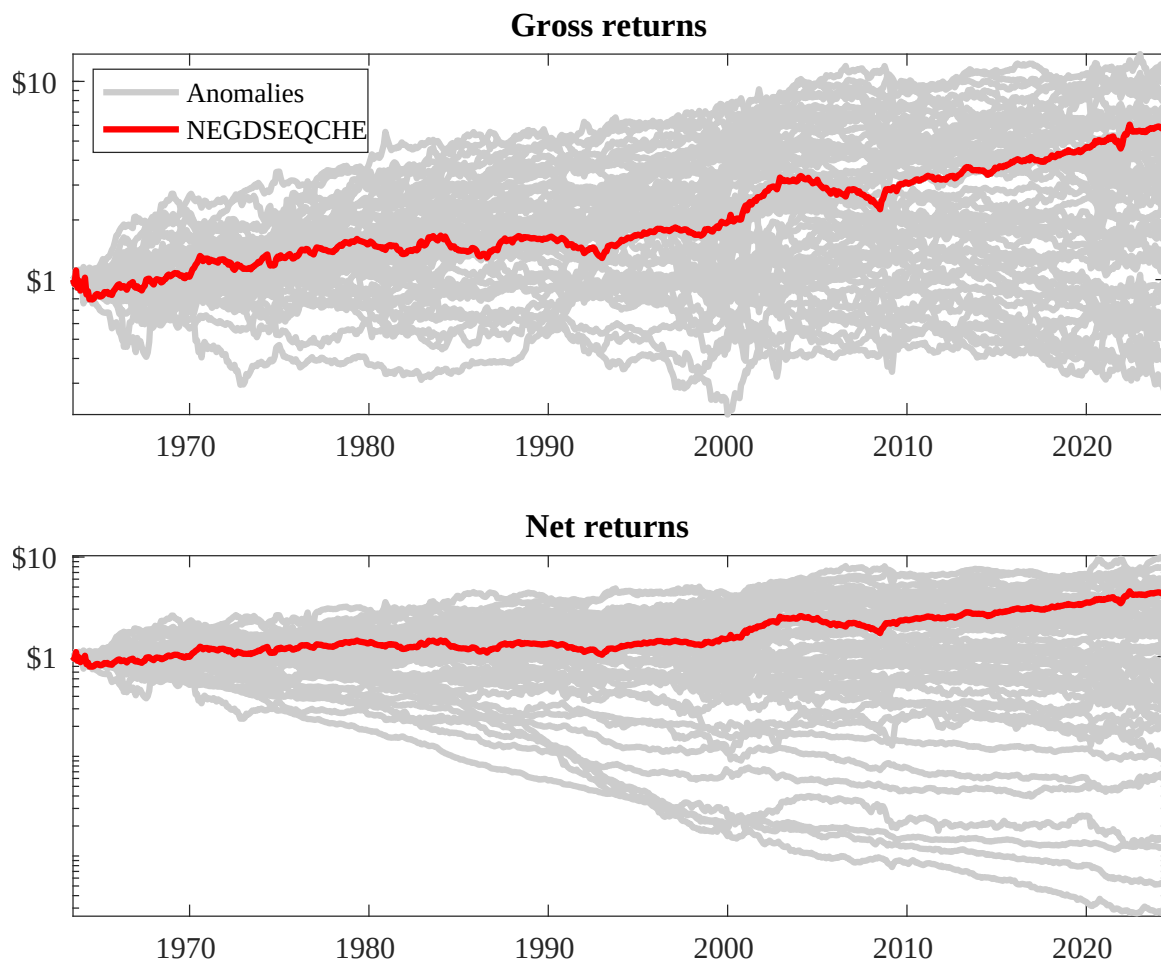
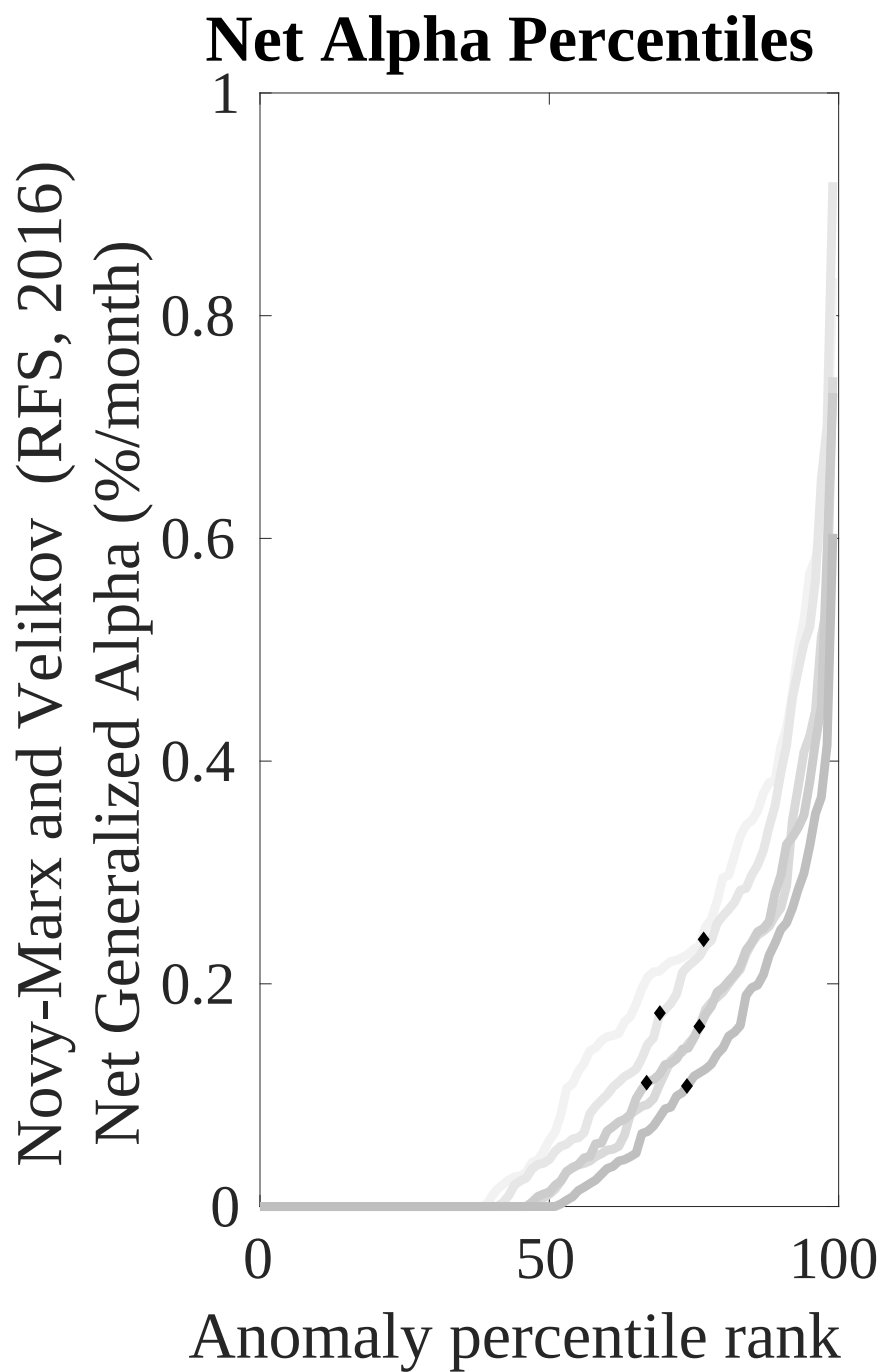
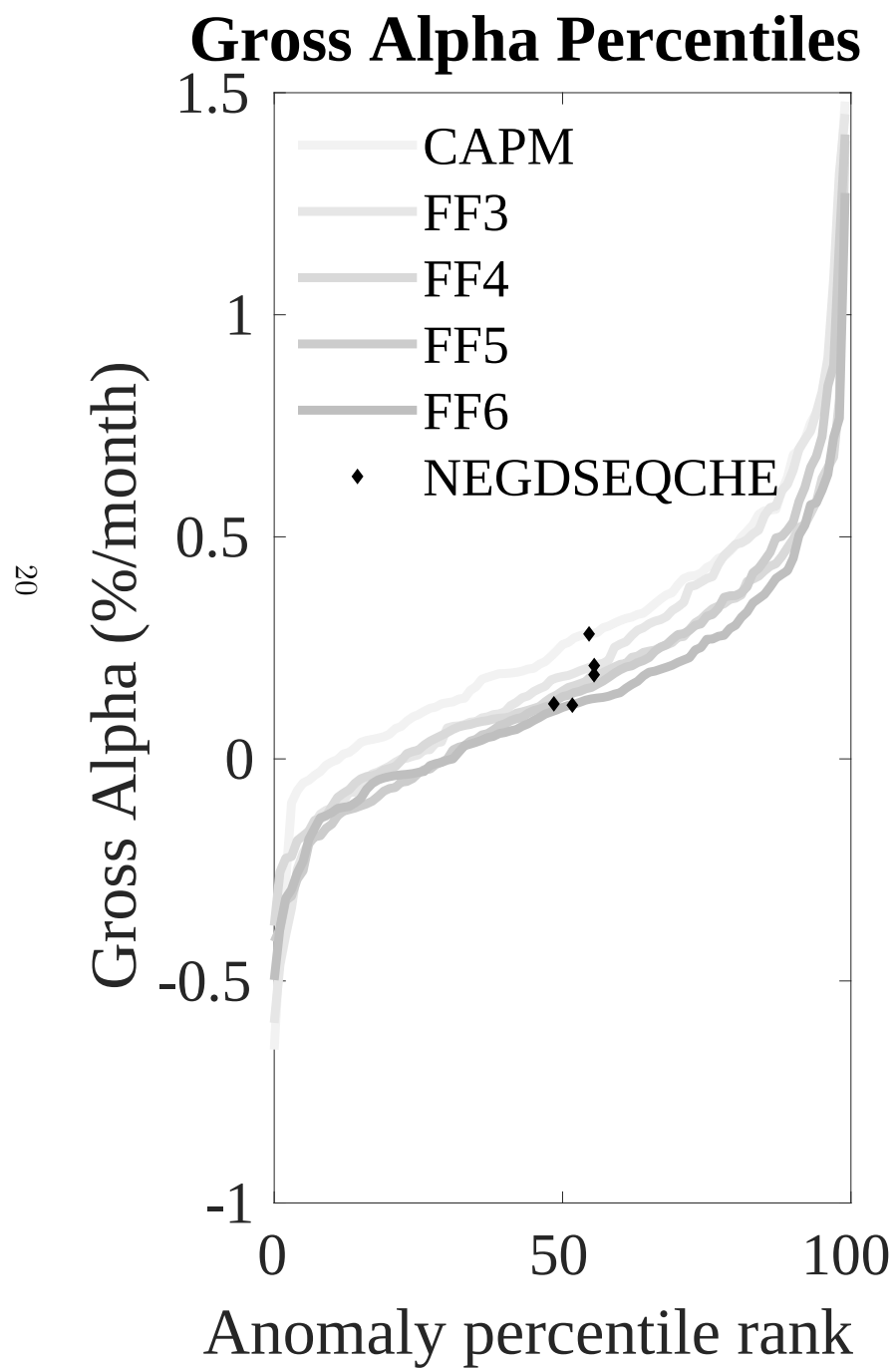


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EDI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



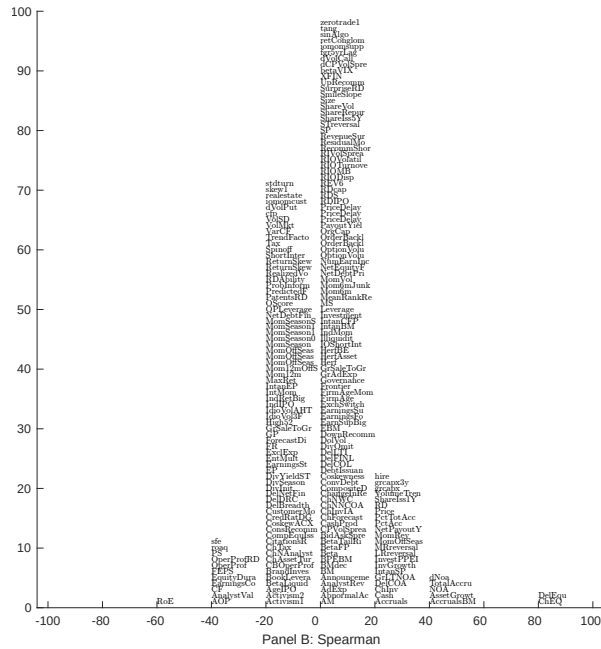
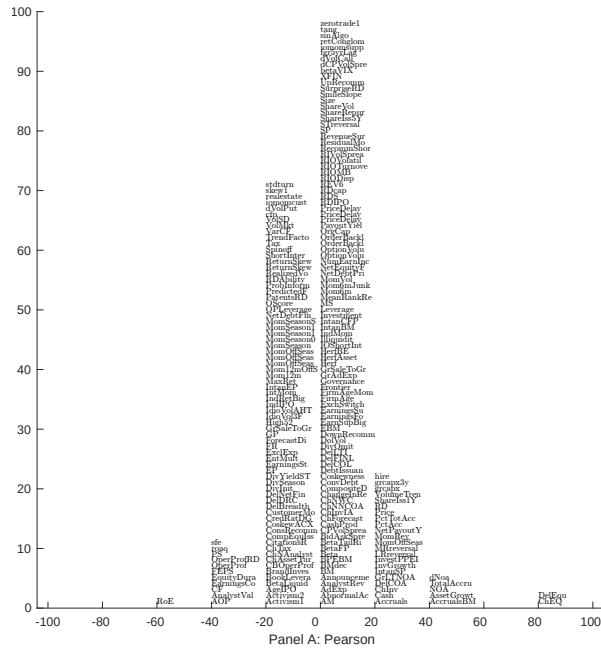


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with EDI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

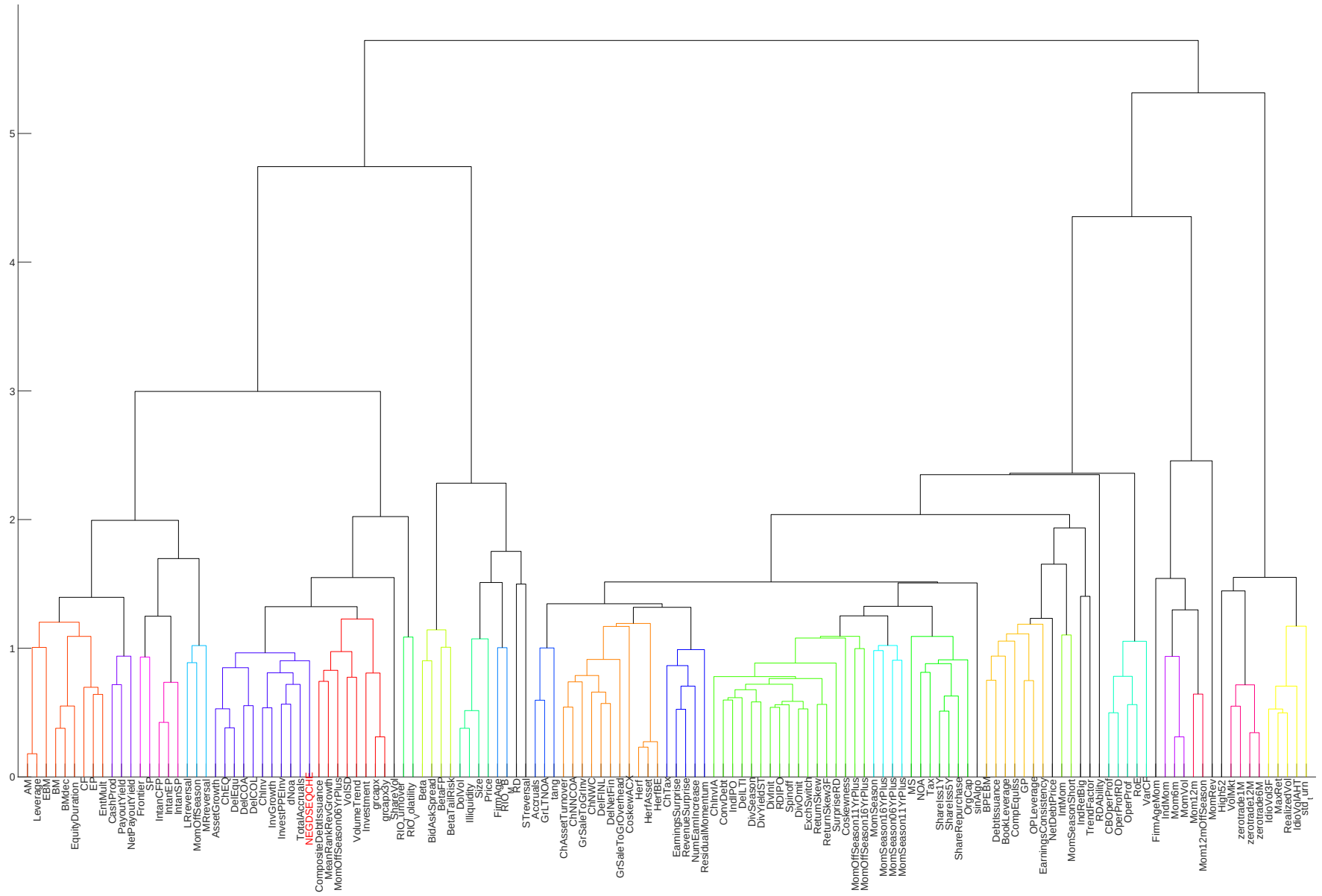


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

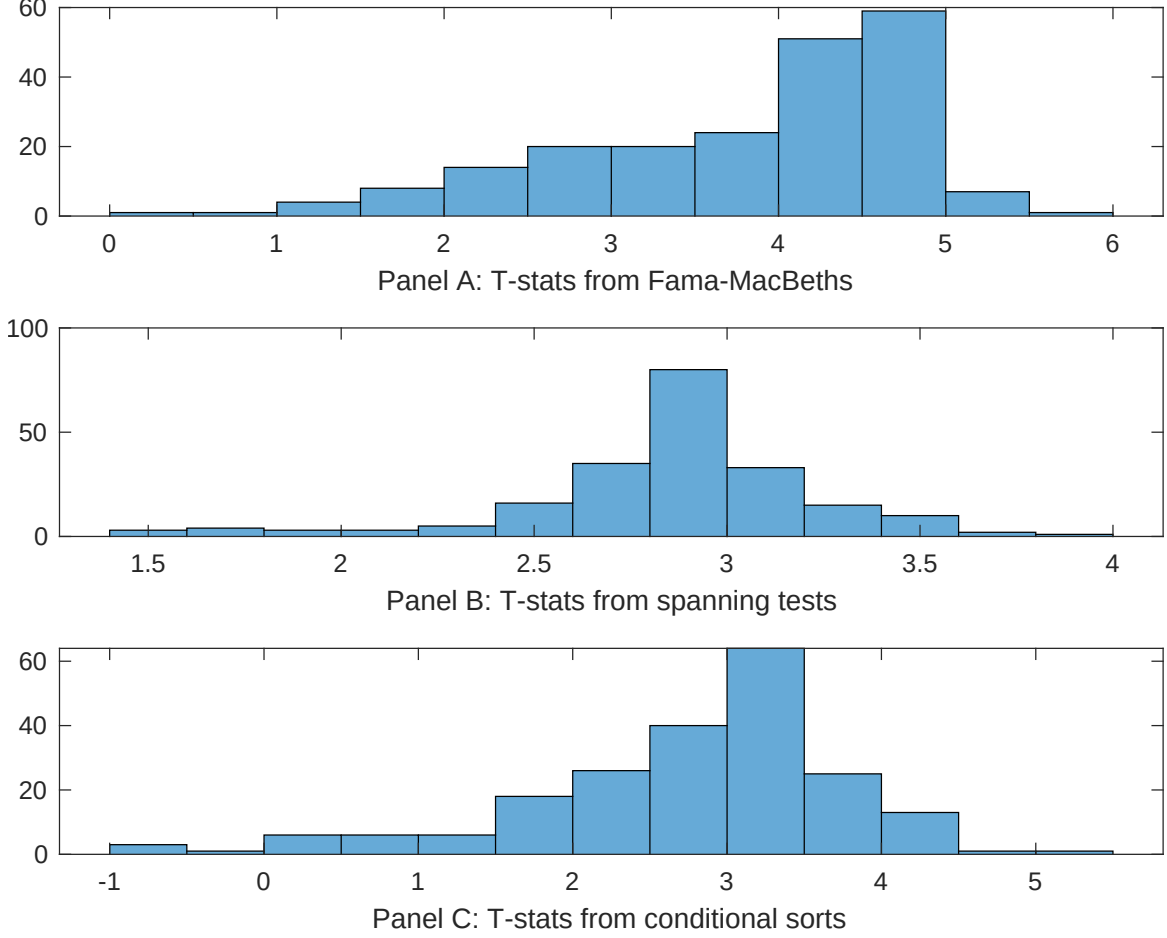


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EDI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDI}EDI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EDI. Stocks are finally grouped into five EDI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EDI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EDI}EDI_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in equity to assets, Growth in book equity, Asset growth, change in net operating assets, Total accruals, Percent Total Accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [5.67]	0.17 [7.39]	0.13 [6.10]	0.13 [5.94]	0.12 [5.40]	0.11 [3.92]	0.12 [5.05]
EDI	0.14 [3.15]	0.16 [3.60]	0.10 [2.20]	0.13 [2.52]	0.24 [4.43]	0.12 [3.34]	0.36 [1.15]
Anomaly 1	0.14 [4.00]						0.62 [1.50]
Anomaly 2		0.44 [4.31]					0.12 [0.13]
Anomaly 3			0.99 [9.29]				0.48 [3.73]
Anomaly 4				0.13 [9.91]			0.41 [2.36]
Anomaly 5					0.34 [1.82]		-0.25 [-1.02]
Anomaly 6						0.13 [3.11]	0.59 [1.37]
# months	708	708	708	708	708	432	432
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EDI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EDI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in equity to assets, Growth in book equity, Asset growth, change in net operating assets, Total accruals, Percent Total Accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [1.48]	0.10 [1.15]	0.12 [1.36]	0.08 [0.87]	0.09 [1.03]	0.26 [2.87]	0.23 [2.82]
Anomaly 1	41.89 [9.60]						10.17 [1.24]
Anomaly 2		47.92 [10.11]					47.47 [6.50]
Anomaly 3			23.29 [4.24]				-21.05 [-3.59]
Anomaly 4				29.57 [5.77]			13.79 [2.67]
Anomaly 5					20.50 [4.85]		-1.40 [-0.27]
Anomaly 6						19.94 [4.95]	11.17 [2.70]
mkt	5.67 [2.70]	8.01 [3.84]	6.47 [2.94]	6.58 [3.02]	5.73 [2.61]	-0.55 [-0.24]	1.25 [0.61]
smb	5.49 [1.81]	4.42 [1.46]	4.16 [1.30]	6.34 [2.02]	6.11 [1.93]	4.84 [1.48]	2.99 [1.01]
hml	-14.74 [-3.60]	-15.02 [-3.69]	-9.91 [-2.33]	-11.46 [-2.71]	-8.96 [-2.12]	-5.97 [-1.49]	-9.73 [-2.61]
rmw	-5.61 [-1.37]	-7.19 [-1.77]	-9.78 [-2.28]	-9.20 [-2.17]	-6.21 [-1.44]	-13.58 [-3.25]	-17.18 [-4.52]
cma	20.34 [2.75]	16.08 [2.15]	34.42 [3.78]	40.47 [5.57]	52.23 [7.95]	51.70 [8.67]	15.36 [1.99]
umd	1.46 [0.70]	-0.38 [-0.18]	1.21 [0.55]	-0.13 [-0.06]	0.98 [0.45]	0.61 [0.30]	-1.85 [-0.99]
# months	731	731	731	731	731	438	438
$\bar{R}^2(\%)$	27	28	19	21	20	31	46

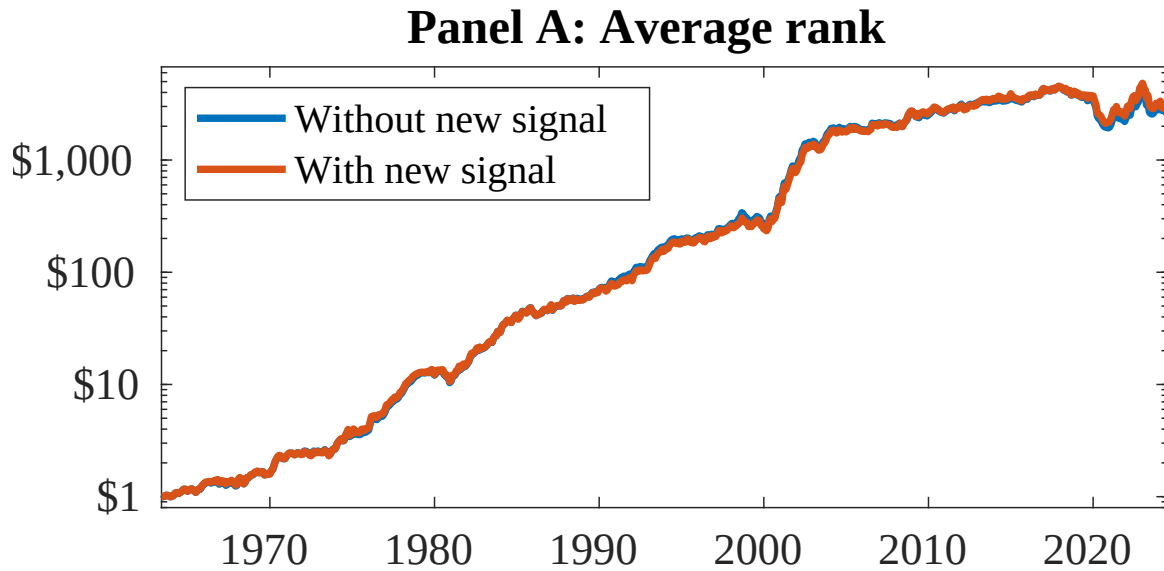


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EDI. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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